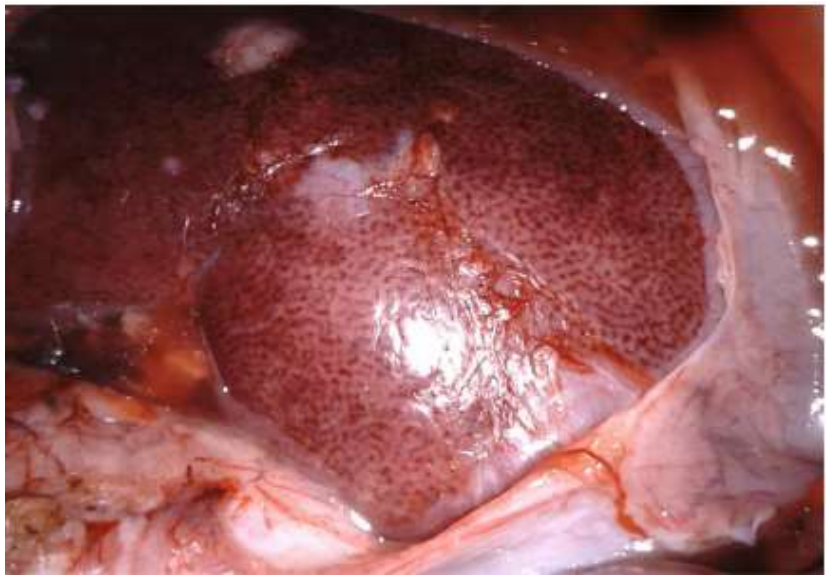
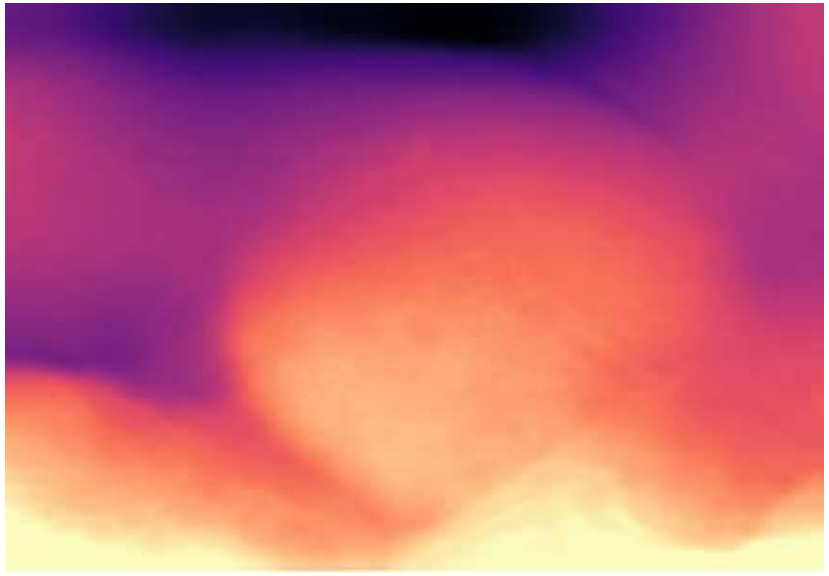
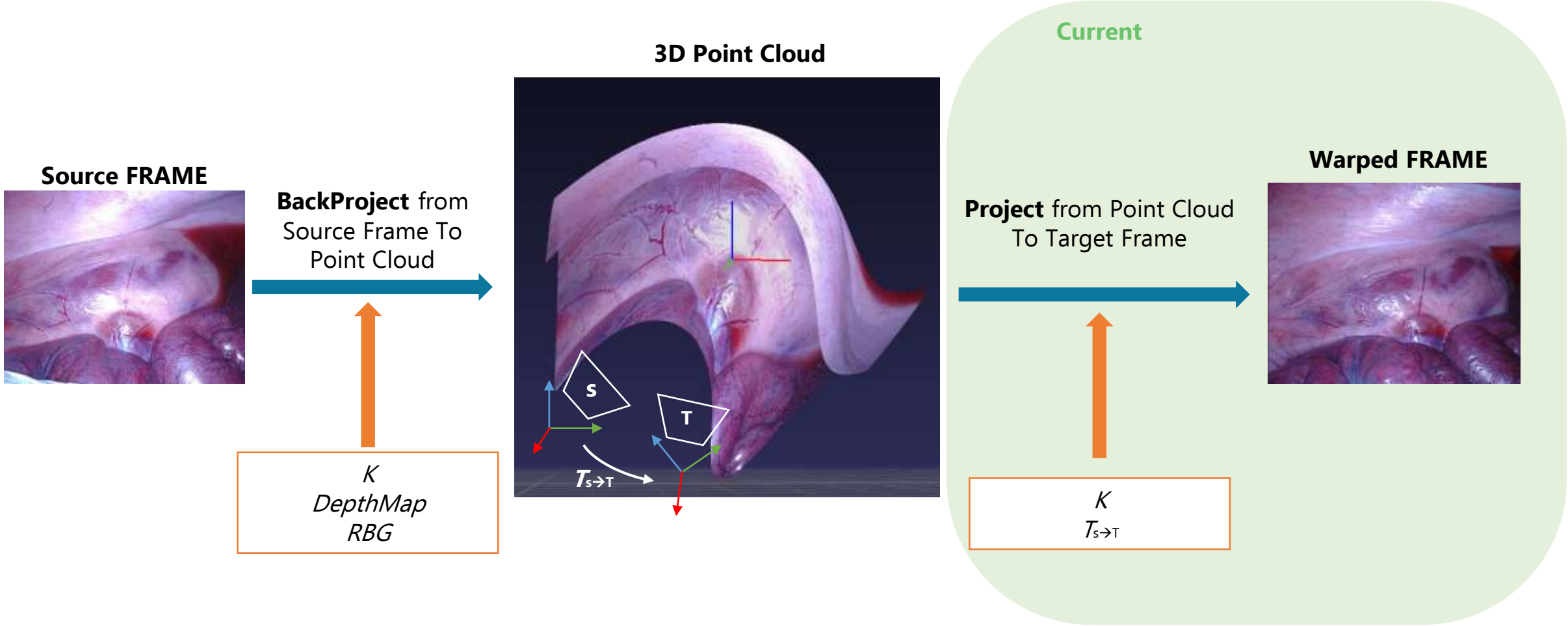
Single-Image Depth Estimation

Projections

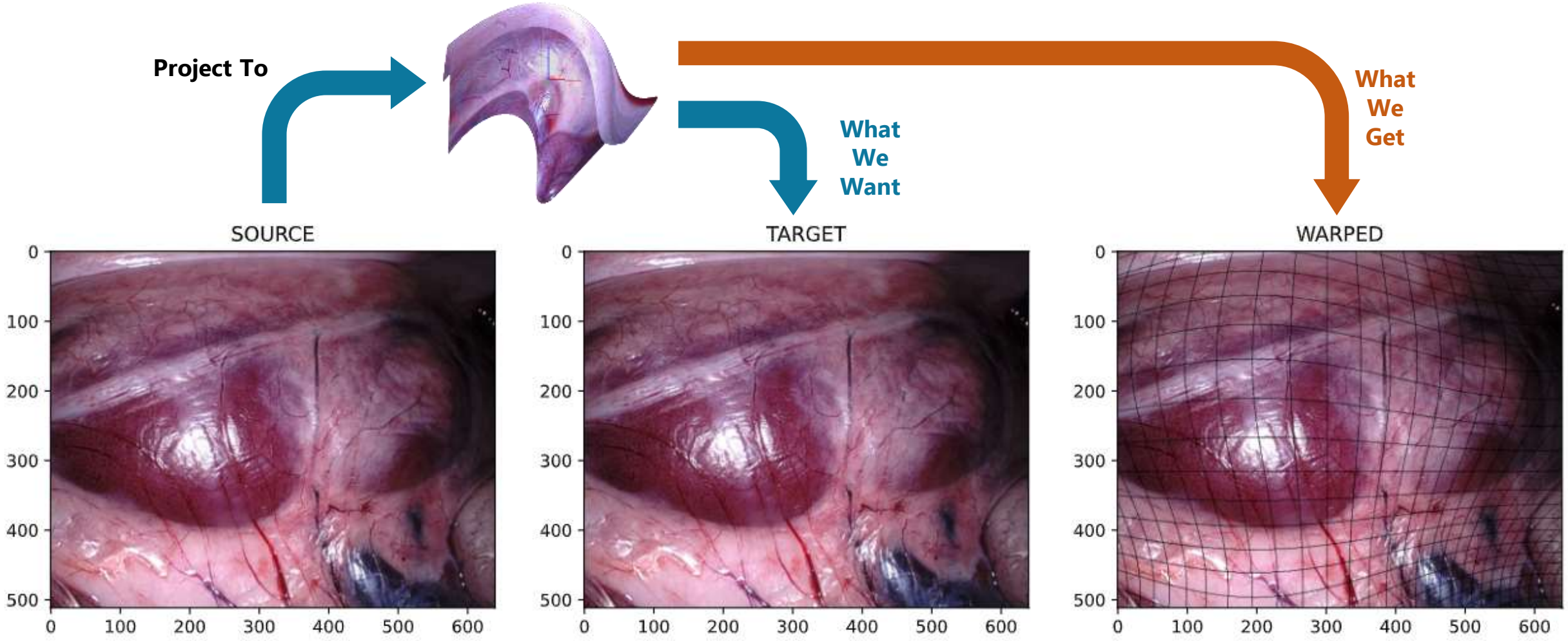




Pipeline


Current
3D Point Cloud
Source FRAME
BackProject from Source Frame To Point Cloud
K
DepthMap
RBG
S
T
$T_{S\rightarrow T}$
Project from Point Cloud To Target Frame
K
$T_{S\rightarrow T}$
Warped FRAME

Projection Results



Artifacts

We obtain **prominent artifacts** during the projection phase

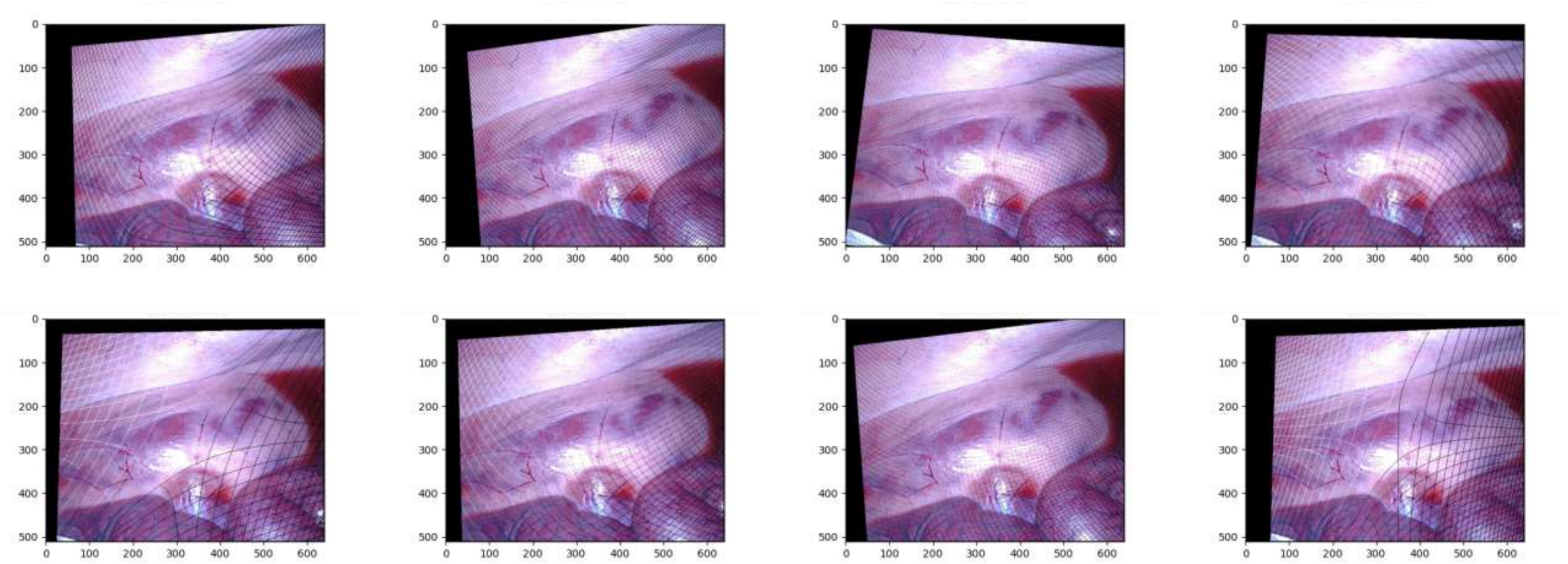
→ Can be removed artificially (*e.g.* median filtering) but it will affect the training phase

Artifacts arise both if:

- We give the **r** and **t** parameters to the *cv.projectPoint* method
- We give **r=(0,0,0)** and **t=(0,0,0)** and transform ourself via matrix multiplication (extrisic)

Artifacts arise at all rotation / translation values

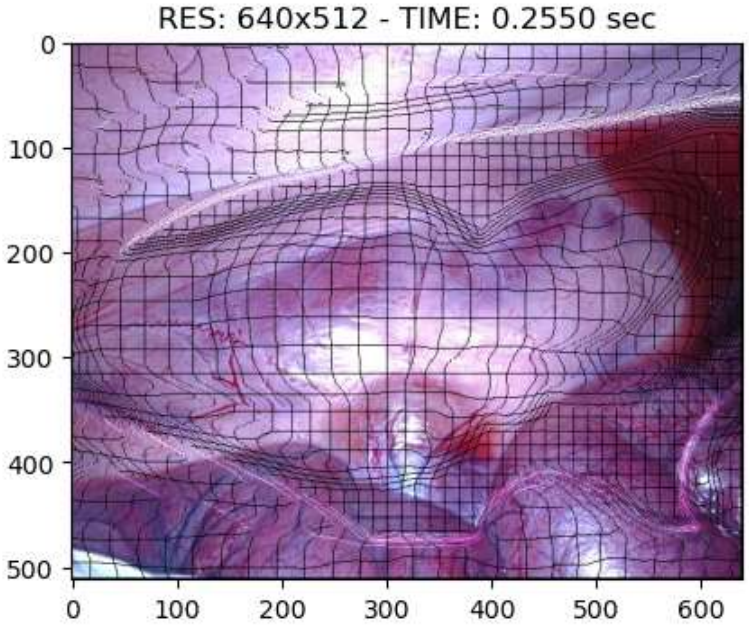
Digging deep…



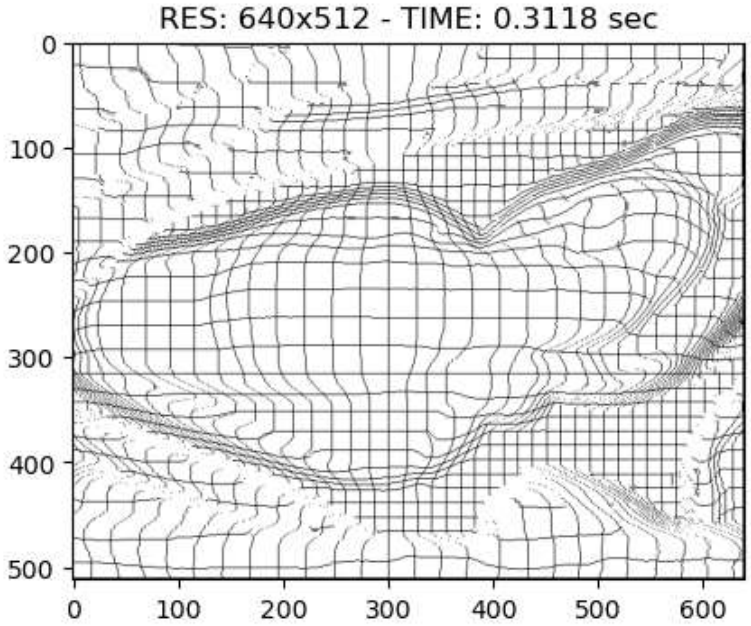
Digging deeper…

Investigation on different causes…

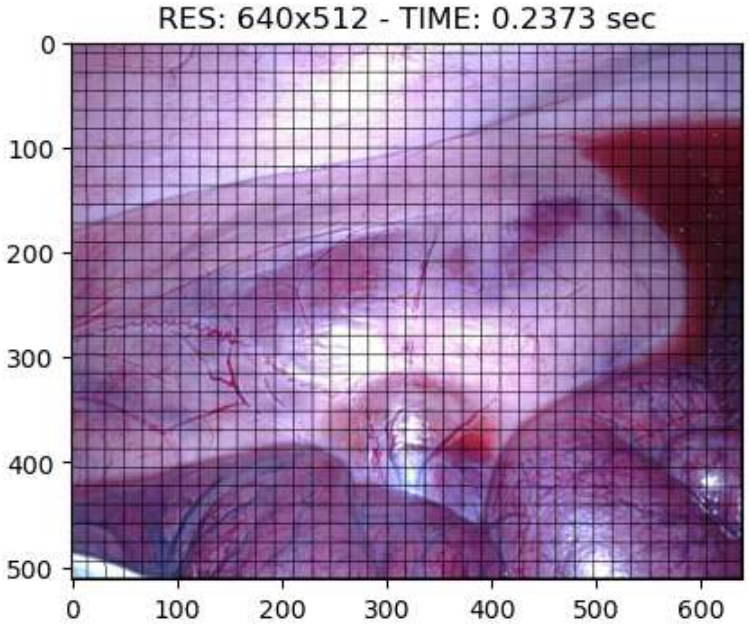
Non-smooth depth map



No color information

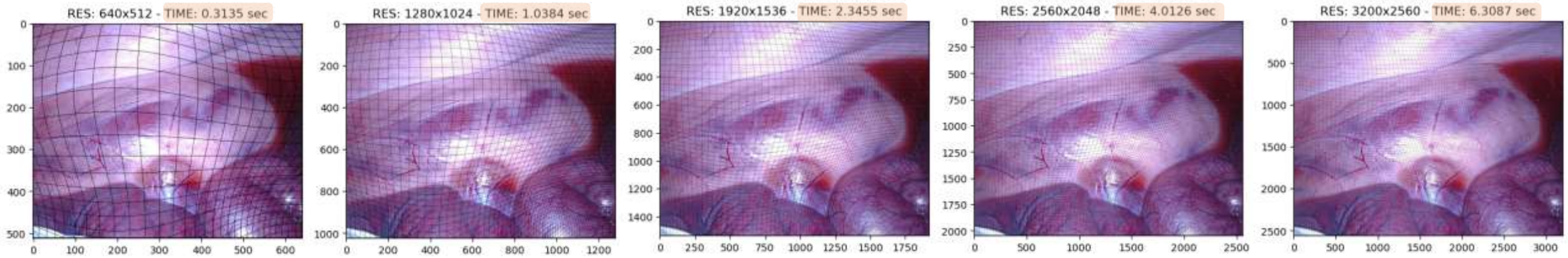


Flat depth map



Aliasing Hypothesis

By (massively, 5x…) increasing the resolution of the projected image, we finally obtain an **artifacts-free** image



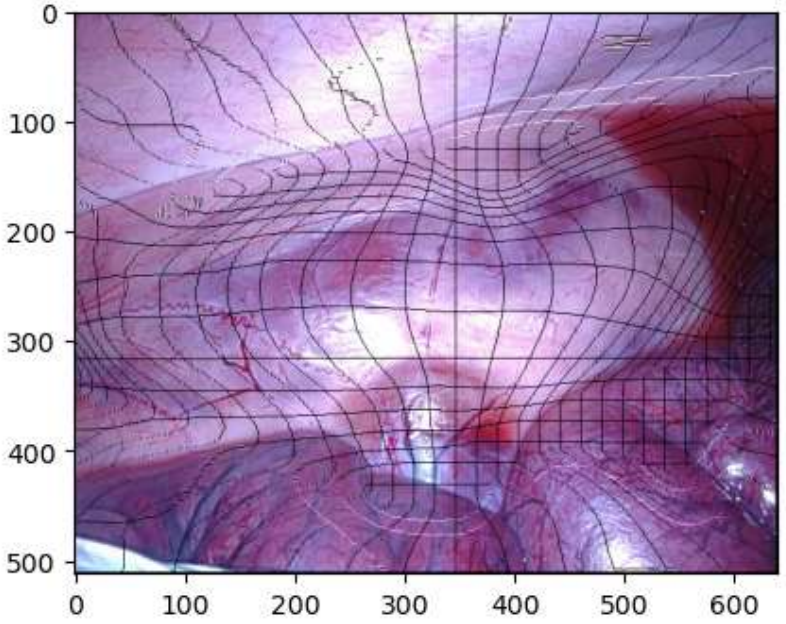
Resolution increases quadratically = Aliasing disappears
Projection time increases $O(n^2)$

Depth map Scaling

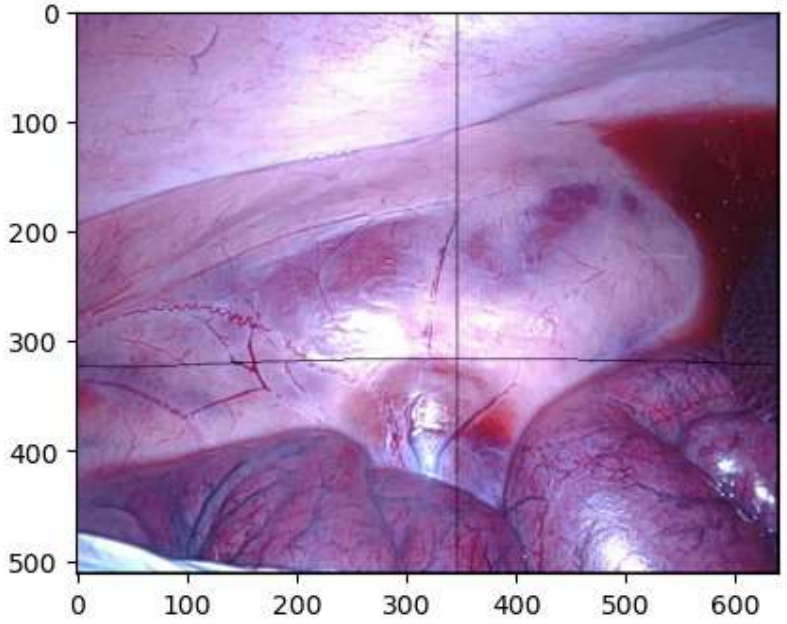
Until now: Depth Range = **4cm – 30cm**

→ Rescaling the depth range → Almost no artifacts

4 – 30



400 – 3’000

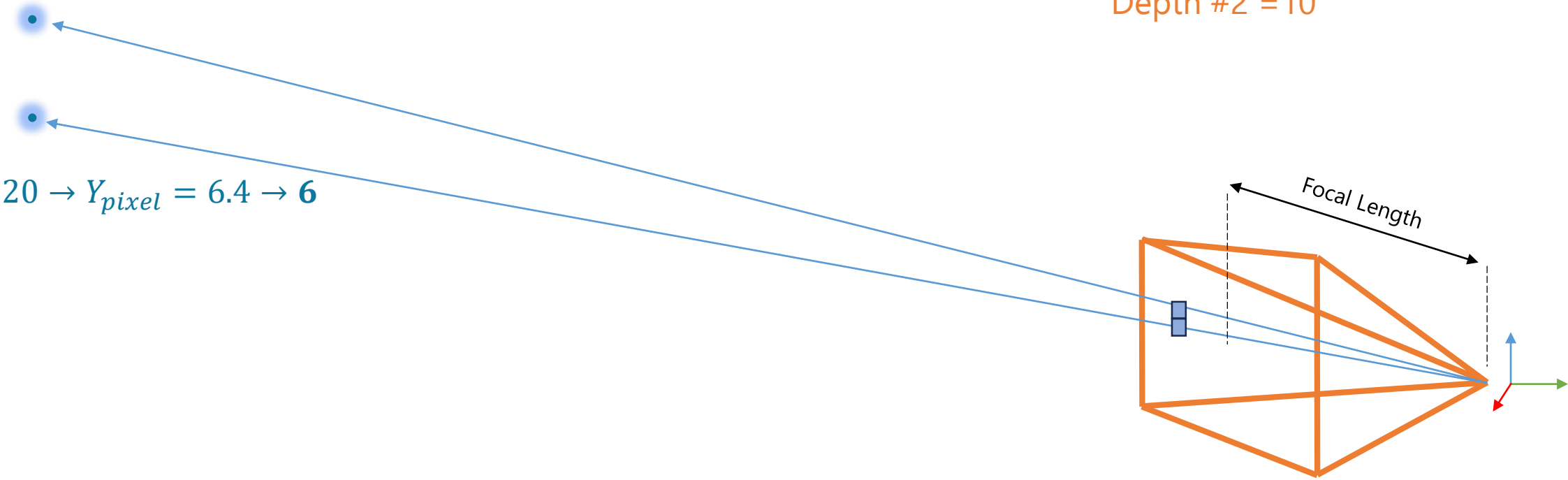


Searching in

Focal Length = 517
Depth #1 = 10'000
Depth #2 =10

$Y_{cloud} = 140 \rightarrow Y_{pixel} = 7.2 \rightarrow \mathbf{7}$

$Y_{cloud} = 120 \rightarrow Y_{pixel} = 6.4 \rightarrow \mathbf{6}$



Digging deeper…

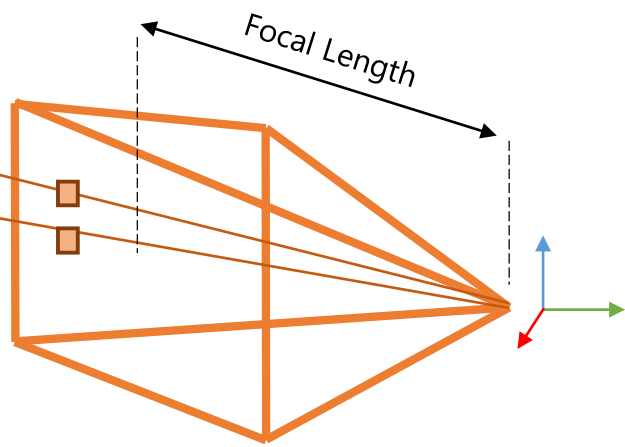
Focal Length = 517
Depth #1 = 10
Depth #2 =10'000

$Y_{cloud} = 140 \rightarrow Y_{pixel} = 7.2 \rightarrow \mathbf{7}$

$Y_{cloud} = 120 \rightarrow Y_{pixel} = 6.4 \rightarrow \mathbf{6}$

$Y_{cloud} = 0.14 \rightarrow Y_{pixel} = 7.5 \rightarrow \mathbf{8}$

$Y_{cloud} = 0.12 \rightarrow Y_{pixel} = 6.4 \rightarrow \mathbf{6}$



Digging deeper…

Focal Length = 517
Depth #1 = 10
Depth #2 =10'000

$Y_{cloud} = 140 \rightarrow Y_{pixel} = 7.2 \rightarrow \mathbf{7}$

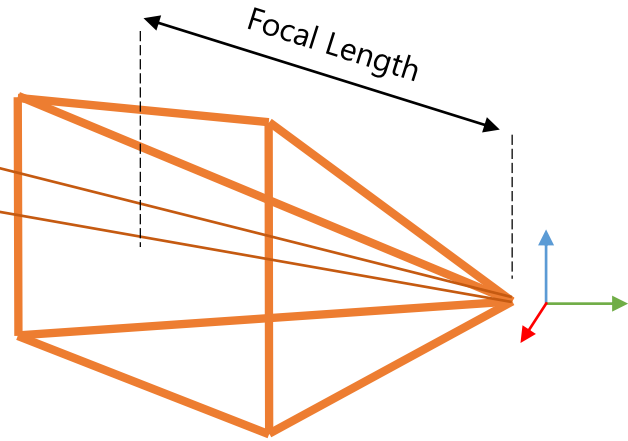
$Y_{cloud} = 120 \rightarrow Y_{pixel} = 6.4 \rightarrow \mathbf{6}$

No Pixel Gap

$Y_{cloud} = 0.14 \rightarrow Y_{pixel} = 7.5 \rightarrow \mathbf{8}$

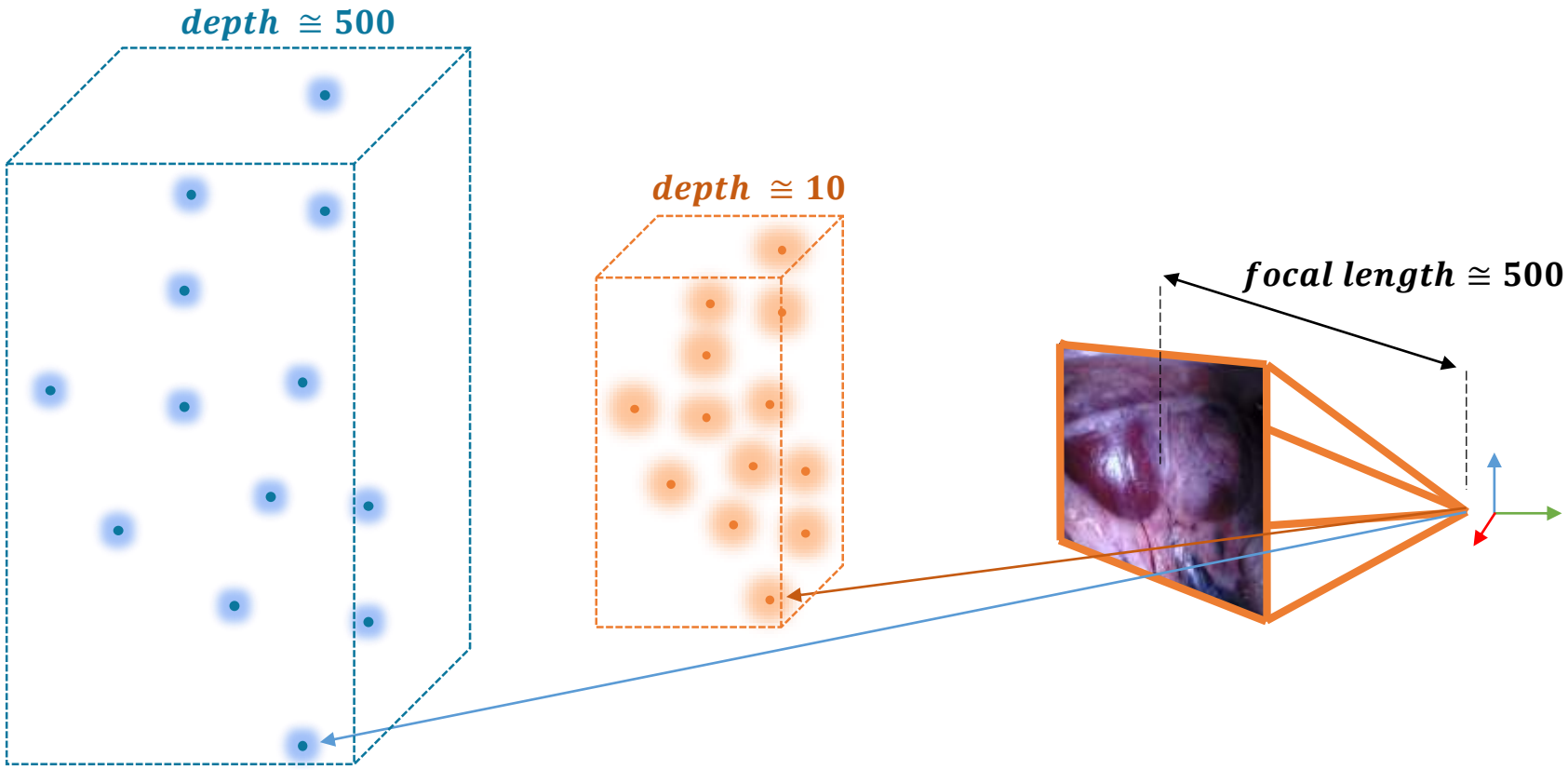
$Y_{cloud} = 0.12 \rightarrow Y_{pixel} = 6.4 \rightarrow \mathbf{6}$

1 pixel **GAP** → Pixel 7 gets no projection

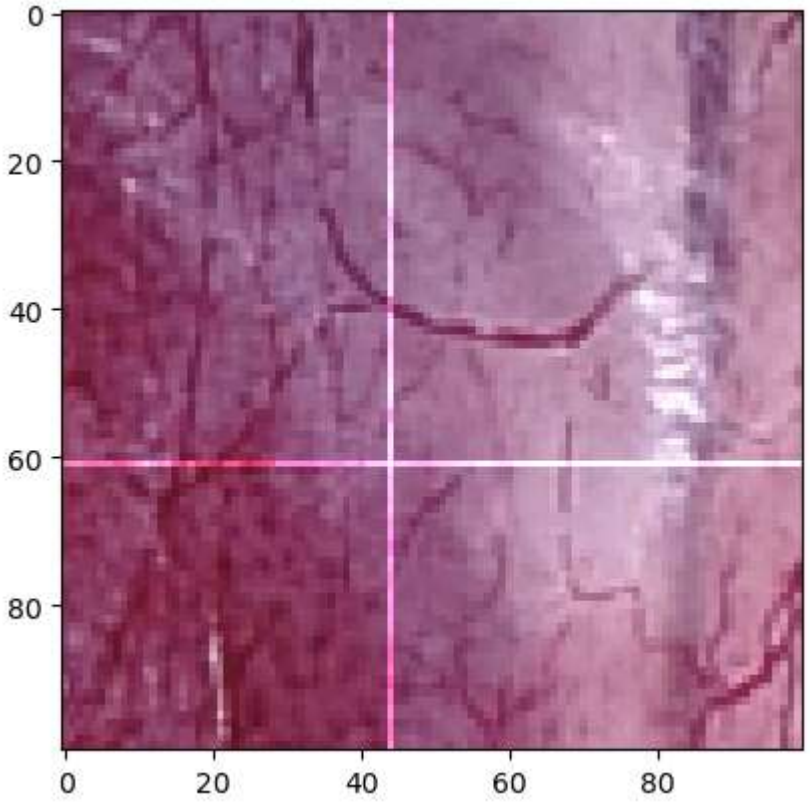


In conclusion

The dimensions of the point cloud must be **comparable** to the focal length



The artifact




```
INTRINSICS =
    tensor([
        [517.587,  0.0000,   344.1809,  0.0000],
        [ 0.0000,  517.489,  260.5354,  0.0000],
        [ 0.0000,  0.0000,   1.0000,    0.0000],
        [ 0.0000,  0.0000,   0.0000,    1.0000]
        ])
```

Next Steps

Next Week:

- Consistently **project artifacts-free** images and build the **loss function**
- **Start a training session**

(Dataset and model architecture are ready)

By the end of September:

- Augmentation
- Some meaningful results